Supplementary Material

Cluster Analysis of CART-Synthesised Data: A Factorial Simulation of k-Means and Ward's Linkage

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Abstract

Supporting figures and tables for the main article, the bootstrap null by stratum, the clustering objectives and the computational cost of the study. Unless stated otherwise, the number of groups is selected over the fixed candidate range $\{2, \ldots, 8\}$, as in the main article.

S1 Recovery of the number of groups by cluster count and separation

Figure S1 presents the recovery rate of the true number of groups as a function of the true cluster count k and the separation σ, for both algorithms. It complements panel (a) of Figure 3 of the main article in a different layout, making the $k \times \sigma$ interaction directly legible: recovery collapses in the lower-left quadrant (many groups, little separation) and is high in the upper-right.

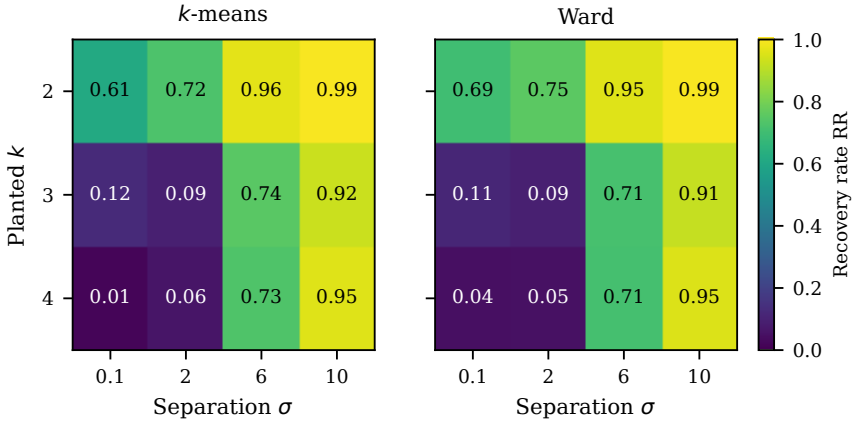


Fig. S1 Recovery rate of the true number of groups (share of synthetic datasets with $\hat{k}_S = k$) by true cluster count and separation, for k-means (left) and Ward (right). Cell values are proportions over the 60,000 evaluations of each cell; candidate range $\{2, \ldots, 8\}$.

S2 Agreement in the estimated number of groups between original and synthetic data

Table S1 cross-tabulates the number of groups estimated from the original data against the number estimated from the corresponding synthetic data, stratified by separation. Unlike the recovery rate, which scores $\hat{k}$ against the construction-time ground truth, this table compares the two estimates with each other and so removes the difficulty of recovering the planted structure; it still contains the ordinary resampling instability of $\hat{k}$, which Section S4 quantifies.

The pattern confirms the expectation that at high separation the mass concentrates on the diagonal: exact agreement rises from 68.4% at $\sigma = 0.1$ to 98.2% at $\sigma = 10$ for k-means (Ward: 64.6% to 97.5%). In the overlapping regime the off-diagonal mass is not symmetric. At $\sigma = 0.1$ the synthetic data yield a *lower* estimate than the original in 15.2% of k-means comparisons and a higher one in 16.4%; the largest single off-diagonal cell is the original data supporting $\hat{k} \geq 5$ while the synthetic data support $\hat{k} = 2$ (8.4%). That is, when the original data contain weak, diffuse structure that the silhouette criterion resolves into many small groups, CART synthesis tends to smooth it away into a single coarse split. The tendency disappears once the groups are genuinely separated.

S3 Sensitivity to the candidate range for the number of groups

The main article selects $\hat{k}$ over the fixed range $\{2, \ldots, 8\}$. Earlier work used $\{2, \ldots, \min(k+3, 8)\}$, a rule that ties the search range to the planted number of groups and so passes information about the true k to the estimator. Table S2 repeats the count analyses under that capped range, applied identically to original and synthetic data,

Table S1 Agreement in the estimated number of groups between original and synthetic data, by separation. Rows are $\hat{k}$ from the original data, columns $\hat{k}$ from the synthetic data; cells are percentages of all comparisons at that separation (each block sums to 100%). Estimates of five or more groups are pooled for display, but the agreement column is computed on unpooled values. Both data sources are searched over the fixed range $\hat{k} \in \{2, \ldots, 8\}$ used in the main article; under the capped range $\{2, \ldots, \min(k+3, 8)\}$ of Section S3, agreement is higher in the overlapping regime (75.1% and 67.6% at $\sigma = 0.1$, 79.7% and 75.9% at $\sigma = 2$, for k-means and Ward respectively) and essentially unchanged at $\sigma \geq 6$, where the two conventions differ by at most 0.1 percentage points. $n = 180{,}000$ comparisons per separation level.

		$\hat{k}$ recovered from synthetic data (%)				Exact
σ	$\hat{k}$ (original)	2	3	4	≥ 5	agreement
k-means						
0.1	2	50.9	2.1	0.5	3.8	68.4%
	3	2.2	10.0	0.5	4.0	
	4	0.0	0.0	0.0	0.0	
	≥ 5	8.4	1.1	0.6	16.0	
2	2	62.9	3.4	1.6	4.2	76.4%
	3	1.2	8.6	0.7	2.2	
	4	0.3	0.4	0.9	1.2	
	≥ 5	2.5	0.9	0.9	8.0	
6	2	38.1	0.9	0.3	0.1	92.6%
	3	0.7	29.1	1.1	0.1	
	4	0.5	2.4	24.7	0.8	
	≥ 5	0.0	0.1	0.2	0.8	
10	2	34.8	0.2	0.2	0.4	98.2%
	3	0.2	31.7	0.3	0.0	
	4	0.0	0.2	31.7	0.3	
	≥ 5	0.0	0.0	0.0	0.0	
Ward						
0.1	2	58.3	5.0	2.0	6.9	64.6%
	3	3.6	4.6	1.1	3.5	
	4	1.2	0.9	0.4	0.9	
	≥ 5	6.3	1.3	0.6	3.4	
2	2	65.7	4.7	2.1	5.3	73.8%
	3	1.1	6.4	1.0	2.1	
	4	1.1	0.5	0.4	1.4	
	≥ 5	2.6	1.0	0.8	3.9	
6	2	37.1	1.0	0.2	0.1	88.2%
	3	1.9	27.3	1.6	0.3	
	4	1.8	2.7	23.6	1.3	
	≥ 5	0.3	0.5	0.1	0.3	
10	2	34.8	0.6	0.1	0.0	97.5%
	3	0.3	31.1	0.3	0.0	
	4	0.0	0.3	31.6	0.4	
	≥ 5	0.4	0.0	0.0	0.2	

and Table S3 repeats the bootstrap null. Narrowing the range raises both agreement measures in the overlapping regime, where the silhouette criterion occasionally selects large values of $\hat{k}$, and leaves them essentially unchanged at $\sigma \geq 6$. The qualitative conclusions are unaffected: the two algorithms remain indistinguishable against the planted number of groups, k-means agrees more often with the original estimate, the advantage arises on Gamma data (0.833 versus 0.749) and not on Normal data (0.896 versus 0.898), and it is matched by the bootstrap null (0.891 versus 0.841). Under the capped range the difference between the two algorithms in the excess over the null is not significant (-0.009, $p = 0.204$). All Wilcoxon statistics in this material and in the main article treat scenarios with exactly equal means as ties (differences are rounded to 10^{-12} before ranking, so that floating-point noise does not break genuine ties).

Table S2 Recovery of the planted number of groups (RR) and agreement with the number estimated from the original data (AG), by separation and distribution family, under the fixed candidate range of the main article and under the capped range. Wilcoxon W and p: paired signed-rank test of k-means against Ward across the 144 scenario-level means. Last row: overall bootstrap null AG_0 ($B = 30$) under each range.

	$\hat{k} \in \{2, \ldots, 8\}$ (main text)				$\hat{k} \in \{2, \ldots, \min(k+3, 8)\}$			
	RR		AG		RR		AG	
Stratum	k-m.	Ward	k-m.	Ward	k-m.	Ward	k-m.	Ward
$\sigma = 0.1$	0.247	0.281	0.684	0.646	0.272	0.296	0.751	0.676
$\sigma = 2$	0.288	0.295	0.764	0.738	0.304	0.307	0.797	0.759
$\sigma = 6$	0.810	0.789	0.926	0.882	0.810	0.789	0.926	0.882
$\sigma = 10$	0.954	0.950	0.982	0.975	0.954	0.950	0.983	0.976
Normal	0.593	0.604	0.869	0.890	0.604	0.610	0.896	0.898
Gamma	0.557	0.554	0.809	0.731	0.565	0.561	0.833	0.749
Overall	0.575	0.579	0.839	0.811	0.585	0.585	0.864	0.823
Wilcoxon W	3508.5		2088.5		3294.5		1665.5	
Wilcoxon p	0.750		5.4×10^{-5}		0.380		2.7×10^{-7}	
AG_0 (bootstrap)			0.871	0.826			0.891	0.841

S4 Bootstrap null and independent-pairs baseline

To judge whether a given level of agreement is high, it helps to know how often a dataset that differs from the original only by resampling yields the same estimate. The main article uses a bootstrap null: each of the 720 original datasets is resampled $B = 30$ times with replacement, the scaler fitted on the original data is applied unchanged, the same silhouette scan is run, and AG_0 is the share of resamples whose estimate equals $\hat{k}_R$. Because proper synthesis in `synthpop` fits its trees to a bootstrap sample of the original data, the resample is the synthesis procedure without the tree step. Table S4 sets AG and AG_0 beside a second, coarser reference retained from the earlier analysis: the proportion of the 10 pairs of independent original datasets in a scenario

Table S3 Bootstrap null under the capped candidate range $\{2, \ldots, \min(k+3, 8)\}$: agreement after CART synthesis (AG), after bootstrap resampling (AG_0, $B = 30$) and the excess $AG - AG_0$ (computed before rounding). Compare with Table 3 of the main article.

		k-means			Ward		
Stratum	Level	AG	AG_0	Excess	AG	AG_0	Excess
Separation σ	0.1	0.751	0.775	−0.024	0.676	0.681	−0.005
	2	0.797	0.840	−0.042	0.759	0.775	−0.016
	6	0.926	0.960	−0.034	0.882	0.923	−0.040
	10	0.983	0.990	−0.007	0.976	0.984	−0.008
Distribution F	Normal	0.896	0.910	−0.014	0.898	0.911	−0.013
	Gamma	0.833	0.872	−0.039	0.749	0.770	−0.021
Normal, σ	0.1	0.765	0.766	−0.001	0.806	0.807	−0.001
	2	0.888	0.911	−0.023	0.878	0.889	−0.010
	6	0.947	0.974	−0.028	0.924	0.964	−0.040
	10	0.984	0.988	−0.004	0.983	0.986	−0.003
Gamma, σ	0.1	0.737	0.783	−0.046	0.545	0.554	−0.009
	2	0.707	0.769	−0.062	0.639	0.661	−0.021
	6	0.906	0.946	−0.040	0.841	0.881	−0.041
	10	0.981	0.991	−0.010	0.969	0.983	−0.014
Overall		0.864	0.891	−0.027	0.823	0.841	−0.017

that yield the same $\hat{k}_R$. The centroids are redrawn for every original dataset (only the minimum pairwise centroid distance is fixed at σ), so two such datasets differ in their centroid geometry as well as in the sample. The pairs baseline is therefore not a second sample from the same population and rests on only 10 pairs per scenario; it is reported as a secondary reference. Both references agree that most of the disagreement in the overlapping regime is present without synthesis, but only the bootstrap null holds the geometry fixed.

S5 Geometric floor and scale-free metrics by stratum

Tables S5 and S6 report, for every geometric metric, the observed value after CART synthesis, the bootstrap floor (the same metric computed at the planted k between the partition of the original data and the partition of a bootstrap resample, averaged over the $B = 30$ resamples) and the excess of the observed value over the floor. MCD, MVD and ΔG are defined in the main article; MVD/p divides MVD by the number of coordinates; $\text{MCD}_{\text{sd}} = \text{MCD}/s_w$, where $s_w = \{\sum_i \|\mathbf{x}_i - \boldsymbol{\mu}_{g_i}\|^2/(Np)\}^{1/2}$ is the pooled within-planted-group standard deviation of the original dataset in the scaled coordinates; and $\text{RVD} = k^{-1}\sum_j |v_j^S/v_j^R - 1|$ is the relative variance difference over the aligned clusters. The observed scale-free metrics are computed for all 720,000 original–synthetic pairs, and the observed MCD, MVD and ΔG recomputed in the same pass coincide with the stored values. Two patterns stand out. The excess of RVD grows with separation for both algorithms, whereas absolute MVD declines, so the decline

Table S4 Synthetic-versus-original agreement (AG), the bootstrap null (AG_0, $B = 30$ resamples per original dataset) and the independent-pairs baseline (Pairs: proportion of pairs of independent original datasets in the same scenario with equal $\hat{k}_R$, which also differ in centroid geometry). Means over scenarios; candidate range $\{2, \ldots, 8\}$.

		k-means			Ward		
Stratum	Level	AG	AG_0	Pairs	AG	AG_0	Pairs
Separation σ	0.1	0.684	0.721	0.792	0.646	0.644	0.667
	2	0.764	0.814	0.714	0.738	0.754	0.722
	6	0.926	0.960	0.764	0.882	0.923	0.758
	10	0.982	0.989	0.933	0.975	0.984	0.922
Distribution F	Normal	0.869	0.880	0.853	0.890	0.902	0.869
	Gamma	0.809	0.863	0.749	0.731	0.750	0.665
Overall		0.839	0.871	0.801	0.811	0.826	0.767

of MVD is a decline of the floor and of the units. The observed values are lower for k-means in every stratum, but the excess is not: overall it is larger for k-means for MCD and MCD_{sd} and equal for the other metrics, with no consistent ordering between the algorithms across strata.

S6 Collapse cells

At low separation the silhouette scan often returns the smallest candidate, $\hat{k} = 2$, on the original and the synthetic data alike, so that the two estimates agree for a reason unrelated to the preservation of structure. We call a scenario a *collapse cell* if all five of its original datasets give $\hat{k}_R = 2$ (with $n = 5$, the threshold of at least 90% means all five). Because $\hat{k} = 2$ is the correct answer when $k = 2$, the substantive subset is $k \geq 3$ (96 scenarios, 480 original datasets). Table S7 reports the share of these original datasets with $\hat{k}_R = 2$, the number of collapse cells, the share of all 720,000 evaluations that fall inside them, the agreement inside and outside them, and, over all k, the share of original datasets with $\hat{k}_R = 2$ at $p = 10$ and $\sigma \leq 2$. All collapse cells lie at $\sigma \leq 2$.

S7 Clustering objectives

k-means partitions N observations $\{\mathbf{x}_i\}_{i=1}^N$, $\mathbf{x}_i \in \mathbb{R}^p$, into k non-overlapping groups by minimising the within-cluster sum of squares (MacQueen, 1967). Writing $\mathcal{C} = \{C_1, \ldots, C_k\}$ for the partition and $\boldsymbol{\mu}_j$ for the centroid of C_j,

$$\arg\min_{\mathcal{C}} \sum_{j=1}^{k} \sum_{\mathbf{x}_i \in C_j} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2, \qquad \boldsymbol{\mu}_j = \frac{1}{|C_j|} \sum_{\mathbf{x}_i \in C_j} \mathbf{x}_i. \tag{S1}$$

Lloyd's iteration alternates assignment and update steps until the partition stabilises, converging to a local optimum.

Table S5 Geometric floor by stratum for the metrics of the main article: observed value after CART synthesis (Obs.), bootstrap floor ($B = 30$) and excess (Obs. minus floor), averaged over the scenario means of each stratum, at the planted k. Strata: separation (36 scenarios each), distribution family (72 each) and dimensionality (48 each).

		k-means			Ward		
Metric	Stratum	Obs.	Floor	Excess	Obs.	Floor	Excess
MCD	$\sigma = 0.1$	0.604	0.497	0.107	0.859	0.801	0.058
	$\sigma = 2$	0.382	0.259	0.123	0.706	0.639	0.068
	$\sigma = 6$	0.136	0.077	0.059	0.177	0.115	0.062
	$\sigma = 10$	0.090	0.054	0.036	0.095	0.057	0.038
	Normal	0.289	0.209	0.080	0.424	0.373	0.051
	Gamma	0.317	0.235	0.082	0.495	0.433	0.062
	$p = 2$	0.141	0.107	0.035	0.272	0.252	0.020
	$p = 5$	0.314	0.237	0.077	0.478	0.428	0.050
	$p = 10$	0.454	0.323	0.131	0.628	0.528	0.100
	Overall	0.303	0.222	0.081	0.460	0.403	0.057
MVD	$\sigma = 0.1$	0.353	0.276	0.077	0.613	0.557	0.057
	$\sigma = 2$	0.240	0.169	0.072	0.535	0.483	0.052
	$\sigma = 6$	0.171	0.095	0.076	0.285	0.188	0.098
	$\sigma = 10$	0.112	0.059	0.052	0.133	0.068	0.065
	Normal	0.147	0.089	0.057	0.277	0.215	0.063
	Gamma	0.291	0.211	0.081	0.506	0.433	0.073
	$p = 2$	0.064	0.047	0.017	0.134	0.122	0.012
	$p = 5$	0.200	0.144	0.056	0.388	0.340	0.049
	$p = 10$	0.393	0.259	0.134	0.653	0.509	0.143
	Overall	0.219	0.150	0.069	0.392	0.324	0.068
ΔG	$\sigma = 0.1$	0.035	0.027	0.008	0.093	0.090	0.003
	$\sigma = 2$	0.032	0.023	0.008	0.086	0.080	0.005
	$\sigma = 6$	0.026	0.016	0.009	0.035	0.023	0.012
	$\sigma = 10$	0.028	0.019	0.009	0.030	0.020	0.010
	Normal	0.028	0.020	0.008	0.057	0.050	0.007
	Gamma	0.032	0.023	0.010	0.065	0.057	0.008
	$p = 2$	0.029	0.021	0.008	0.056	0.050	0.006
	$p = 5$	0.030	0.022	0.008	0.060	0.053	0.007
	$p = 10$	0.031	0.022	0.009	0.067	0.057	0.010
	Overall	0.030	0.021	0.009	0.061	0.053	0.008

Hierarchical clustering builds a dendrogram by successive merging (Ward, 1963). Each observation begins as a singleton; at every step the two clusters whose merger least increases the total within-cluster error sum of squares are combined. With

$$\mathrm{ESS} = \sum_{j} \sum_{\mathbf{x}_i \in C_j} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2, \qquad \Delta_{\mathrm{Ward}}(C_a, C_b) = \frac{|C_a|\,|C_b|}{|C_a| + |C_b|}\,\|\boldsymbol{\mu}_a - \boldsymbol{\mu}_b\|^2, \tag{S2}$$

a flat partition into k groups is obtained by cutting the dendrogram at the corresponding height. The two methods therefore minimise the same within-cluster sum of squares, k-means by iterative reassignment and Ward's linkage by greedy, irreversible merging.

S8 Computational complexity and cost

This section states what the study costs to run and why. Throughout, N is the sample size, p the dimensionality, k the number of groups, I the number of Lloyd iterations

Table S6 Scale-free metrics by stratum: MVD per coordinate (MVD/p), centroid distance in within-component standard deviations (MCD_{sd}) and relative variance difference (RVD). Columns and strata as in Table S5. Observed values use all 720,000 original–synthetic pairs.

		k-means			Ward		
Metric	Stratum	Obs.	Floor	Excess	Obs.	Floor	Excess
MVD/p	$\sigma = 0.1$	0.065	0.050	0.015	0.119	0.108	0.011
	$\sigma = 2$	0.042	0.030	0.012	0.099	0.091	0.008
	$\sigma = 6$	0.025	0.015	0.010	0.042	0.029	0.013
	$\sigma = 10$	0.016	0.009	0.007	0.020	0.011	0.008
	Normal	0.025	0.016	0.009	0.052	0.044	0.009
	Gamma	0.050	0.036	0.013	0.088	0.076	0.011
	$p = 2$	0.032	0.024	0.008	0.067	0.061	0.006
	$p = 5$	0.040	0.029	0.011	0.078	0.068	0.010
	$p = 10$	0.039	0.026	0.013	0.065	0.051	0.014
	Overall	0.037	0.026	0.011	0.070	0.060	0.010
MCD_{sd}	$\sigma = 0.1$	0.605	0.498	0.107	0.860	0.802	0.058
	$\sigma = 2$	0.442	0.302	0.141	0.820	0.743	0.077
	$\sigma = 6$	0.217	0.125	0.092	0.279	0.182	0.098
	$\sigma = 10$	0.189	0.115	0.074	0.201	0.121	0.080
	Normal	0.349	0.247	0.103	0.502	0.428	0.074
	Gamma	0.377	0.273	0.104	0.578	0.496	0.082
	$p = 2$	0.194	0.143	0.050	0.349	0.316	0.033
	$p = 5$	0.375	0.276	0.099	0.559	0.488	0.071
	$p = 10$	0.522	0.361	0.161	0.712	0.581	0.131
	Overall	0.363	0.260	0.103	0.540	0.462	0.078
RVD	$\sigma = 0.1$	0.093	0.072	0.022	0.194	0.179	0.015
	$\sigma = 2$	0.075	0.054	0.021	0.190	0.177	0.013
	$\sigma = 6$	0.077	0.047	0.030	0.125	0.090	0.036
	$\sigma = 10$	0.084	0.048	0.035	0.099	0.058	0.041
	Normal	0.062	0.038	0.024	0.117	0.092	0.024
	Gamma	0.102	0.072	0.030	0.188	0.160	0.028
	$p = 2$	0.099	0.071	0.028	0.201	0.180	0.021
	$p = 5$	0.080	0.054	0.025	0.148	0.122	0.026
	$p = 10$	0.068	0.040	0.027	0.108	0.076	0.031
	Overall	0.082	0.055	0.027	0.152	0.126	0.026

Table S7 Collapse cells for $k \geq 3$ under the fixed candidate range $\{2, \ldots, 8\}$ of the main article and under the capped range. Datasets $\hat{k}_R = 2$: share of the 480 original datasets with $k \geq 3$ for which the silhouette scan selects two groups. Collapse cells: number of the 96 scenarios with $k \geq 3$. Evaluations in cells: share of all 720,000 evaluations. AG inside and outside: agreement in the evaluations with $k \geq 3$ inside and outside the collapse cells. Last column: share of all original datasets (all k) with $\hat{k}_R = 2$ at $p = 10$ and $\sigma \leq 2$.

		Datasets	Collapse	Evaluations	AG		$\hat{k}_R = 2$,
Range	Algorithm	$\hat{k}_R = 2$	cells	in cells	inside	outside	p=10, $\sigma \leq 2$
fixed	k-means	0.348	22	0.153	0.955	0.799	0.808
	Ward	0.406	26	0.181	0.929	0.762	0.900
capped	k-means	0.356	22	0.153	0.966	0.820	0.883
	Ward	0.410	26	0.181	0.933	0.770	0.925

to convergence, $K = 7$ the number of candidate values of $\hat{k}$, and $A = 2$ the number of algorithms.

S8.1 Cost of the components

Table S8 collects the asymptotic cost of each component together with the number of times the executed design invokes it. The invocation counts follow from the design: 144 scenarios $\times$ $n = 5$ replicates $= 720$ units, each with $m = 1000$ synthetic draws, giving 720,000 original–synthetic pairs.

Table S8 Cost per invocation and invocation count. Exact minimisation of the k-means objective is NP-hard in general and remains so for $k = 2$ in arbitrary dimension (Inaba, Katoh, and Imai, 1994), so Lloyd's local-search heuristic (Lloyd, 1982) is used, restarted ten times. Sources: CART (Breiman, Friedman, Olshen, and Stone, 1984); Lloyd iteration (MacQueen, 1967; Lloyd, 1982); Ward linkage (Ward, 1963; Murtagh and Contreras, 2017); silhouette (Rousseeuw, 1987); Hungarian alignment (Kuhn, 1955; Jonker and Volgenant, 1987).

Component	Cost per invocation	Invocations
CART synthesis	$O(p^2 N \log N)$	7.2×10^5
Lloyd iteration	$O(IkNp)$ per restart ($\times 10$)	5.77×10^6
Ward linkage	$O(N^2 p)$ time, $O(N^2)$ space	5.77×10^6
Silhouette	$O(N^2 p)$	1.15×10^7
Hungarian alignment	$O(k^3)$	1.44×10^6

Two structural facts follow. First, the *binding* term is $O(N^2 p)$, contributed by Ward's linkage and by the silhouette, and it is a property of the sample size rather than of the design: doubling N quadruples the work, whereas doubling m or the number of scenarios only doubles it. The $O(N^2)$ working set of the linkage is also what caps N in practice, since each worker holds its own distance structure; at $N = 1000$ this is roughly 8 MB per worker and irrelevant, at $N = 10^4$ it is 800 MB and binding. Second, model selection dominates the fit count: estimating $\hat{k}$ costs K fits and K silhouette evaluations per algorithm per dataset, so the fidelity stage performs $720 \times A(K+1) + 720{,}000 \times A(K+1) = 1.15 \times 10^7$ clustering fits rather than the 1.44×10^6 that fitting at a single known k would require. Where a cheaper formulation exists we did not pursue it: Ward admits an $O(N^2)$ nearest-neighbour-chain implementation (Murtagh and Contreras, 2017; Müllner, 2011) and k-means can be seeded to bound its expected error (Arthur and Vassilvitskii, 2007), so the figures below are conservative relative to a tuned implementation.

One algorithmic choice does change the constant materially. The partition of each original dataset is computed once per replicate and reused across that replicate's 1000 draws. Computing it inside the draw loop would multiply the original-side work by 10^3 and produce identical output, since the original data do not vary within a replicate.

S8.2 Parallel execution

The design is a set of independent units, so the computation is embarrassingly parallel in the strict sense: no communication, no shared mutable state, and a serial fraction confined to reading configuration and writing the merged result. Amdahl's law is therefore not the operative constraint at this scale; hardware contention is, and the implementation targets three specific sources of it.

Thread oversubscription. The clustering routines call into BLAS, which by default spawns one thread per core *inside each worker*. With P worker processes this creates $P \times C$ runnable threads on C cores, and the resulting context-switching and cache thrashing dominate the useful work. All three stages therefore pin `OMP_NUM_THREADS`, `OPENBLAS_NUM_THREADS` and `MKL_NUM_THREADS` to 1 before importing any numerical library, leaving parallelism entirely to the process pool. The effect is not marginal: on the reference machine, 18 concurrent workers each running an identical model-selection scan complete in 1.9 s with threads pinned and 84.5 s without, a factor of 44.

Load imbalance. Unit cost varies with p and k, so a static partition of the task list leaves workers idle at the tail. The synthesis stage instead uses a work-stealing queue: tasks are serialised into a spool directory and each worker claims one by `rename(2)` into a *doing* directory. Rename within a filesystem is atomic, so exactly one worker can win a given task and no lock is required. Workers pull candidates in batches of 200 rather than listing the spool per task, which at 1.44×10^5 tasks reduces directory-scan overhead from hours to seconds. The Python stages achieve the same effect through a dynamically scheduled pool.

Restart cost. Each unit checkpoints its own result on completion, so an interrupted run resumes at the last completed unit rather than from the start. At a stage cost of several hours this converts a failure from a total loss into the loss of the units in flight.

Figure S2 shows the measured strong-scaling curve. The shape is the expected one for a compute-bound, memory-light workload on a simultaneous multithreading processor, and it makes the operating point explicit: 18 workers buy a 10.0× speed-up over serial execution, while 12 would give 8.8× at much better efficiency. Neither choice is a contribution; the point of measuring is that the default of one worker per hardware thread is not obviously right, and here it is not.

S8.3 Measured cost

The synthetic datasets occupy 21 GB as uncompressed Parquet. Cost is linear in m: the same design at $m = 100$ completes in about a tenth of the time, and every marginal mean of Table 2 of the main article (under both candidate ranges) and every overall and marginal mean of MCD, MVD and ΔG differs from the $m = 1000$ value by at most 0.0052, so the larger run buys precision rather than a change in any conclusion.

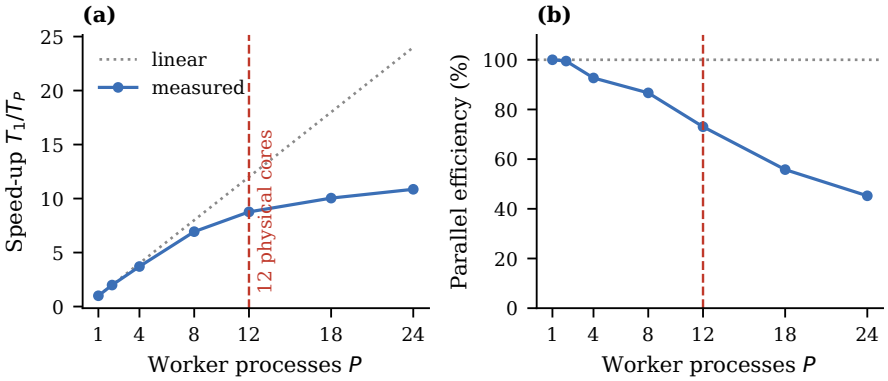


Fig. S2 Strong scaling of the model-selection workload on the reference machine, 72 independent units, BLAS pinned to one thread per worker: (a) speed-up over serial execution, (b) parallel efficiency. The dashed vertical line marks the 12 physical cores. Scaling is near-linear to 8 workers (6.9×, 87% efficiency) and the knee falls at 12, the physical core count. Beyond it the additional workers occupy SMT siblings that share execution units with an already-saturated core, so throughput still rises but efficiency falls to 56% at 18 workers and 45% at 24. The production runs use 18: past the efficiency knee, but chosen for wall time on an otherwise idle machine. At 56% efficiency, the clustering stage costs 3.6 h rather than the 2.0 h that dividing its serial cost by 18 would suggest.

Table S9 Wall-clock cost at $m = 1000$ on an AMD Ryzen 9 9900X (12 cores, 24 threads, 46 GiB) with 18 workers. The evidence differs by row: the fidelity stage is instrumented; synthesis is the interval over which its 1.44×10^5 output files (one per scenario and draw, each holding that draw for all five replicates, 7.2×10^5 synthetic datasets in total) were written, so it includes any idle time inside that interval; the clustering stage was not instrumented, so its entry is a measurement on a uniform random sample of 3000 of its 1.44×10^5 inputs, run at the same worker count and scaled to the full set, which is consistent with the loose upper bound of 5.6 h implied by its output timestamps. The bootstrap null of the main article (Section S4) took a further 363 s on 16 workers, and the scale-free metrics for all 720,000 pairs 745 s.

Stage	Evidence	Wall time	Unit cost
Synthesis (7.2×10^5 datasets)	write interval	1.24 h	6.2 ms/dataset
Clustering (7.2×10^5 pairs)	sampled	≈ 3.6 h	≈ 18 ms/pair
Fidelity (1.15×10^7 fits)	instrumented	4.44 h	1.4 ms/fit
Total		≈ 9.3 h	

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